

Deloitte Full-Stack Developer Preparation Roadmap (12 Months)

Goal:

Become interview-ready for Deloitte T&T Consultant - Innovation Cloud Development Center (Full-stack Developer) and develop advanced software engineering skills.

Learning Strategy:

This roadmap uses spiral learning. Every new phase builds on previous knowledge. Projects continuously reuse old skills so knowledge is not forgotten.

Month 1 - Foundation Layer

Topics:

- Linux fundamentals
- Bash scripting
- Terminal workflow
- Git and GitHub
- Computer science basics
- Processes, memory, networking basics

Project:

Developer Automation Toolkit

Build:

- setup.sh
- backup.sh
- deploy.sh
- cleanup scripts
- Documentation repository

Skills retained:

Linux, Bash, Git, documentation.

Month 2 - Advanced Python

Topics:

- Advanced Python
- OOP
- SOLID principles
- Decorators
- Generators
- Async programming
- Typing
- pytest
- Clean code

Project:

CLI Task Manager

Features:

- User management
- Tasks
- Search
- Export
- Automated tests

Month 3 - Databases and Backend APIs

Topics:

- PostgreSQL
- SQL optimization
- Database design
- FastAPI
- REST APIs
- JWT authentication
- Database migrations

Project:

Professional Task Management API

Stack:

FastAPI + PostgreSQL + Docker + pytest

Features:

- Authentication
- Roles
- CRUD
- Pagination
- Filtering

Month 4 - JavaScript and TypeScript

Topics:

- Modern JavaScript
- ES6+
- Async programming
- DOM
- TypeScript
- Interfaces
- Generics
- Type safety

Project:

Full Stack Task Manager

Stack:

React + TypeScript + FastAPI + PostgreSQL

Month 5 - Node.js and NestJS

Topics:

- Node.js
- Express
- NestJS
- Middleware
- Dependency Injection
- Authentication

Project:
Replace FastAPI backend with NestJS backend.

Goal:
Understand multiple backend ecosystems.

=====

Month 6 - React and Next.js

=====

Topics:

- React architecture
- Hooks
- State management
- Next.js
- SSR
- SSG
- Performance optimization

Project:
Enterprise Dashboard

Features:

- Authentication
- Charts
- Admin panel
- Responsive design

=====

Month 7 - Docker, Redis and CI/CD

=====

Topics:

- Docker
- Docker Compose
- Redis
- Nginx
- GitHub Actions
- Deployment pipelines

Project:
Production Deployment

Add:

- Containers
- Caching
- Automated deployment

=====

Month 8 - Testing and Security

=====

Topics:

- Jest
- Integration testing
- E2E testing
- OWASP Top 10
- JWT security

- CORS
- XSS
- CSRF

Project:
Secure Enterprise Application

Month 9 - Software Architecture

Topics:

- Clean Architecture
- Design Patterns
- SOLID
- Microservices
- API Gateway
- Event-driven architecture

Project:
Microservices Platform

Build:

- User service
- Product service
- Notification service

Month 10 - Algorithms and System Design

Topics:

- Data structures
- Algorithms
- LeetCode practice
- Scalability
- Load balancing
- Caching

Project:
URL Shortener System

Include:

- Redis
- Database
- Analytics

Month 11 - Cloud and Advanced Frontend

Topics:

- AWS/Azure basics
- Monitoring
- D3.js
- Three.js
- Visualization

Project:

AI Analytics Dashboard

Month 12 - Final Capstone and Interview Preparation

Final Project:
Enterprise AI Project Management Platform

Technology:

Frontend:

- React
- Next.js
- TypeScript

Backend:

- NestJS
- FastAPI

Database:

- PostgreSQL
- MongoDB

Infrastructure:

- Docker
- CI/CD
- Redis

Security:

- OWASP practices
- Authentication
- Authorization

Features:

- Project management
- Kanban boards
- Real-time communication
- Analytics
- AI assistant
- Audit logs

Weekly Revision System

Every week:

70% new learning
30% revision

Every Sunday:

- Review previous concepts
- Improve old projects
- Refactor code
- Update documentation

Rule:

Never abandon old projects.
Each project evolves into the next level.

Final Outcome:

After 12 months you will have:

- Advanced backend skills
- Full-stack TypeScript skills
- DevOps experience
- System design knowledge
- Multiple portfolio projects
- Strong Deloitte interview preparation